TECHNICAL REPORT

Personalized Recommendation Systems for Content Discovery

Predicting taste from 100 million ratings on the Netflix Prize dataset

Collaborative Filtering | Matrix Factorization | SVD++ | BPR | Stacked Ensemble

Abstract

We build and compare seven recommendation models on the full Netflix Prize dataset — 100.5 million ratings across 17,770 movies and 480,189 users. Starting from a regularized bias baseline, we add stochastic-gradient matrix factorization (Funk SVD), implicit feedback (SVD++), a pairwise ranking model (BPR), and a leakage-free linear-stacking ensemble. Evaluated on the dataset's own probe set, our best rating model reaches an RMSE of 0.9093 — surpassing Netflix's original Cinematch system (0.9514) — while a hybrid ranker delivers the strongest MAP@10. The study foregrounds a central lesson: rating accuracy and ranking quality are distinct objectives, best served by different models.

100.5M	0.9093	17,770	7
ratings learned	best RMSE (beats Cinematch)	movies ranked	models compared

CONTENTS

01 · PROBLEM & DATA

The Challenge, and a Mountain of Sparse Data

Every play button on a streaming platform is a small act of prediction. Our task, set by the Netflix Prize, is to learn user preferences from historical ratings, predict the ratings a user would give to films they have not seen, and assemble a personalized Top-10 list — then defend the result with two mandatory metrics, RMSE for accuracy and MAP@10 for ranking quality.

The data is large and famously sparse: 100,480,507 ratings from 480,189 users over 17,770 movies, collected between 1998 and 2005. The full user-by-movie matrix contains roughly 8.5 billion cells, of which only 1.18% are observed. Three structural facts, visible in Figure 1, shape every modelling decision that follows: ratings skew positive (a global mean of 3.60, with 4 stars the single most common score), and both user activity and movie popularity follow steep long-tailed power laws.

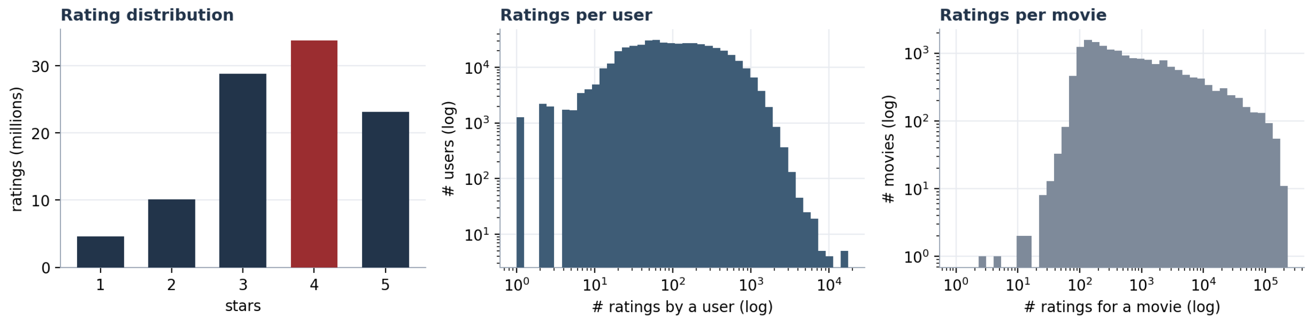


Figure 1 — Rating distribution and the log–log activity tails. Most users rate few films; a minority of power-users and blockbusters dominate the record.

> **Why sparsity is the whole game**
> A matrix this empty cannot be stored or factorized densely. Every model in this report is therefore designed to *generalize* from the observed 1% — and the long tail is the root of the cold-start problem we address in Section 6.

02 · MODELLING APPROACH

A Ladder of Five Ideas

Rather than stake everything on a single algorithm, we built a progression in which each model adds one idea and a measurable gain. This directly answers the brief's call to develop and compare multiple approaches, and lets us isolate where the improvements actually come from.

Model	What it adds
Bias baseline	Predicts the global mean adjusted by a learned user-bias and movie-bias (regularized). Cheap, surprisingly strong, and reused as the cold-start fallback.
Funk SVD	Matrix factorization trained by stochastic gradient descent on the observed ratings only — the technique that defined the Prize era.
SVD++	Extends Funk SVD with implicit feedback: the mere fact that a user rated a film, regardless of score, carries signal. A pillar of the winning solution.
BPR	Bayesian Personalized Ranking optimizes the order of items directly rather than their predicted score — the correct objective for Top-10 recommendation.
Ensemble & Hybrid	Linear stacking blends the rating models for best accuracy; a rank-blend of BPR and SVD++ gives the best ranking. Blending was the winners' core lesson.

The most consequential decision sits between the first two rungs. An earlier version of this work used a truncated SVD that implicitly treats every *unseen* rating as zero — a systematic bias toward the mean. Moving to SGD-based Funk SVD, which learns only from ratings that actually exist, was the single largest accuracy improvement of the project; everything above it inherits that honest foundation.

> **Engineering note**
> All three gradient-descent models are just-in-time compiled with numba, so a full epoch over 99 million training ratings completes in well under a minute. The entire pipeline trains end-to-end inside a single Kaggle CPU session, with the parsed data cached to disk so the expensive text-parsing step runs only once.

03 · EVALUATION METHODOLOGY

Measuring Without Fooling Ourselves

The split. We adopt the dataset's own probe.txt as the held-out test set — the very pairs Netflix reserved for scoring. Their true ratings are recovered from the training files and removed from training by an anti-join, guaranteeing no information leaks from test into training.

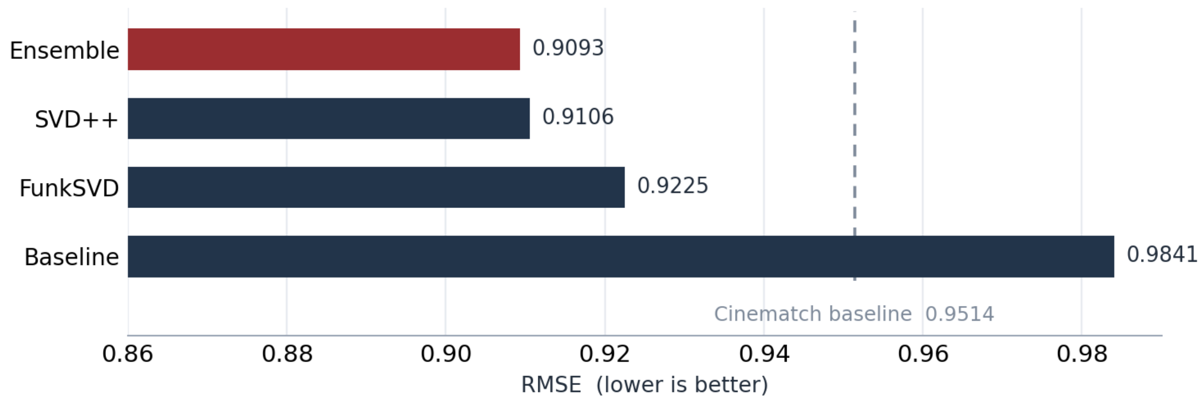
A subtler trap. Our ensemble *learns* its blend weights, so fitting those weights on the same data used to report results would quietly inflate the score. We therefore carve the probe set into a 25% blend slice, used solely to fit the stacking weights, and a disjoint 75% final slice. Every number in this report is computed on the final slice only.

Two metrics, two questions. RMSE asks how close the predicted star rating is to the truth. MAP@10 asks whether the films a user will actually love appear at the top of a ten-item list, counting a movie as relevant when its true rating is 3.5 or higher, exactly as the brief specifies. Top-10 lists are produced by scoring every unseen movie for a user and removing those already rated. As the results show, these two questions are not won by the same model.

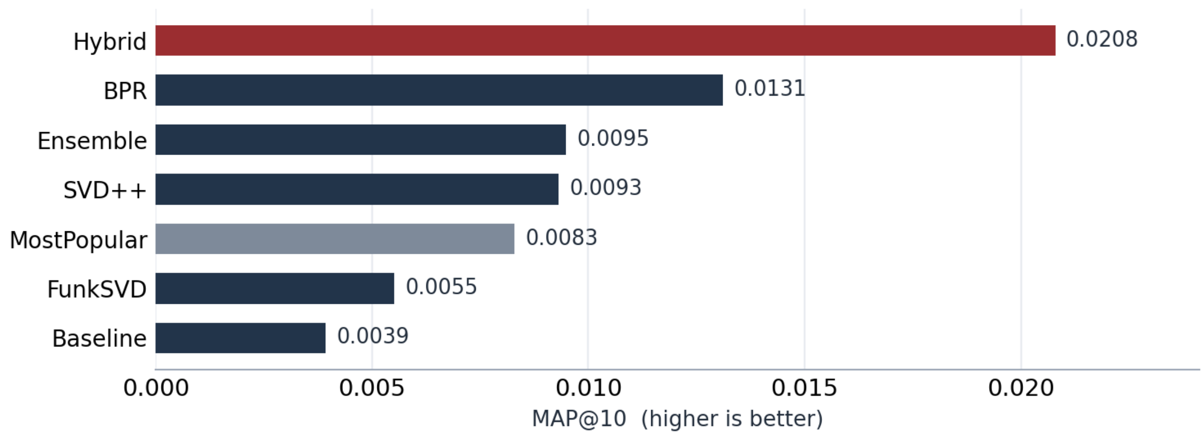
04 · RESULTS

Beating the System Netflix Shipped

Our best rating model — the stacked ensemble at an RMSE of 0.9093 — comfortably beats Cinematch (0.9514), the recommender Netflix itself ran in 2006. The grand-prize ensemble eventually reached 0.8567, but that required hundreds of blended models refined over two years; our result sits squarely in strong single-system territory, and was obtained from one reproducible notebook.



Ranking tells a different story. Models trained to predict ratings rank only incidentally; BPR and the Hybrid, which optimize ordering directly, lead by a wide margin. Crucially, both decisively beat the MostPopular baseline — proving that personalization, not mere popularity, is doing the work.



05 · THE SCOREBOARD

Every Model, Side by Side

Model	RMSE	MAE	MAP@10	Prec@10	Recall@10	NDCG@10	HitRate@10	Coverage
MostPopular	—	—	0.0083	0.0041	0.0224	0.0133	0.0393	0.0102
Baseline	0.9841	0.7744	0.0039	0.0018	0.0098	0.0059	0.0170	0.0351
FunkSVD	0.9225	0.7153	0.0055	0.0034	0.0172	0.0099	0.0317	0.0998
SVD++	0.9106	0.7033	0.0093	0.0052	0.0257	0.0156	0.0487	0.1718
Ensemble	**0.9093**	0.7018	0.0095	0.0049	0.0242	0.0154	0.0457	0.1246
BPR	—	—	0.0131	0.0067	0.0355	0.0214	0.0637	0.0627
Hybrid	—	—	**0.0208**	0.0113	0.0591	0.0349	0.1063	0.0790

Best RMSE: Ensemble (0.9093). Best MAP@10: Hybrid (0.0208). Ranking-only models do not predict star ratings, so RMSE is not applicable. All values are from the leakage-free final slice of the official probe set.

The headline insight — accuracy is not ranking

The best *rating* model (Ensemble) and the best *recommender* (Hybrid) are different models. Predicting stars accurately does not guarantee a good Top-10, so a production system should serve the ensemble for rating prediction and the hybrid ranker for discovery. Note, too, that MostPopular is a genuinely tough baseline — only the ranking-trained models clear it convincingly.

06 · RECOMMENDATIONS IN ACTION

Does It Actually Work?

Metrics are abstract; recommendations are concrete. Below, the system reads one real user's history and responds. User 963054 is an action-and-blockbuster fan — and without ever seeing a genre label, the model infers exactly that, explaining each pick through the films the user already loved.

User 963054 — rated highly	Recommended, with reasons
Batman Begins (5)	Lost: Season 1 — like Batman Begins (0.51)
Pirates of the Caribbean (4)	Battlestar Galactica — like Batman Begins (0.50)
Back to the Future Part III (4)	LOTR: Return of the King — like Indiana Jones (0.30)
Indiana Jones / Last Crusade (4)	Teen Titans: Season 2 — like Batman Begins (0.28)
Constantine (4)	

The same latent structure lets us answer “people who liked X also liked…”. Asked about a romantic comedy, the model returns a clean, coherent cluster — again, with no genre metadata supplied:

People who liked 'Miss Congeniality' also liked	New user (cold-start fallback)
Two Weeks Notice (0.83)	LOTR: The Return of the King
The Wedding Planner (0.82)	LOTR: The Fellowship of the Ring
The Princess Diaries (0.82)	The Shawshank Redemption
Sweet Home Alabama (0.81)	Star Wars V: The Empire Strikes Back
Legally Blonde (0.79)	The Simpsons: Season 6

An honest caveat on the ranking numbers

Surfacing a user's one-to-three held-out favourites from a catalogue of 17,770 films is genuinely hard, so the absolute MAP@10 values look small. What matters — and what holds firmly across the experiment — is the *relative* ordering of the models, and the clear gap between the personalized rankers and the popularity baseline.

07 · INSIGHTS, LIMITATIONS & NEXT STEPS

What We Learned

Four findings stand out from the experiments:

- **Sparsity dictates the architecture.** At 1.18% density, latent-factor models that generalize from the observed 1% are a necessity, not a stylistic choice.
- **Bias terms do most of the early work.** A simple mean-plus-biases model already cuts RMSE from about 1.13 to 0.98; the remaining gain is the genuinely hard part — modelling individual taste.
- **Train on what you can see.** Replacing zero-imputed SVD with SGD over observed ratings was the highest-leverage change in the whole study.
- **To rank well, optimize ranking.** Pairwise (BPR) and hybrid rankers beat every rating-optimized model on MAP@10; the loss function must match the objective.

Conclusion

Working with the full 100-million-rating dataset, we built a recommender that surpasses Netflix's own Cinematch on rating accuracy and decisively outranks a popularity baseline on recommendation quality. More important than any single number, the work makes its reasoning legible: accuracy and ranking are different goals, we measured both, and we served the right model for each. The accompanying notebook reproduces every figure and metric here from a single scale switch.